

HNSS07 · The Brain — Pathology Chart

The brain disorders presented in lecture and the principles of brain dysfunction | **STEP 1** = beyond class content · **EXAM EMPHASIS** = flagged in class as testable

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Brain Disorders Presented in Lecture

Telencephalon		Diencephalon		Midbrain		Pons		Cerebellum	
Broca's aphasia · Huntington		pituitary adenoma		Parkinson disease		pontine stroke		medulloblastoma	
DISORDER	WHO	REGION · STRUCTURE	WHAT GOES WRONG	SYMPTOMS & SIGNS	BUZZWORDS / STEP 1 ADD-ONS	TREATMENT · PROGNOSIS · KEY FACT			
Broca's aphasia <i>the halting speaker</i> TELENCEPHALON	Any cause: stroke (often MCA STEP 1), tumor, trauma — the site matters, not the cause	Broca's area + surrounding tissue (left inferior frontal gyrus STEP 1)	Damage to a language region and its network (classic case: Tan's brain , studied by Paul Broca)	Speech and language impairment; also memory , executive and motor problems	Non-fluent, effortful speech; comprehension largely intact; poor repetition. Contrast Wernicke : fluent nonsense, poor comprehension (superior temporal gyrus) STEP 1	Speech therapy; rehab STEP 1 . One structure → several functions			
Huntington disease <i>the dancing movements</i> TELENCEPHALON	Adults (~30s–50s) STEP 1 ; autosomal dominant	Basal ganglia — caudate nucleus — and other structures	Degeneration of the caudate	Chorea (brief, abrupt, irregular, unpredictable movements) + memory , executive , language deficits	CAG repeat (HTT, chr 4), anticipation; caudate atrophy → enlarged lateral ventricles; ↓ GABA, ↓ ACh, ↑ dopamine STEP 1	Supportive (tetrabenazine STEP 1). Basal ganglia are NOT just motor EXAM EMPHASIS			
Pituitary adenoma <i>the tumor that narrows the view</i> DIENCEPHALON	Common : ~17% prevalence in the general population	Pituitary gland (hypothalamus–pituitary)	Tumor, benign or malignant ; may oversecrete a hormone and/or compress neighbors	Bitemporal hemianopsia (compresses the optic chiasm); Cushing syndrome from excess ACTH → excess adrenal cortisol	Most common type = prolactinoma (galactorrhea, amenorrhea; cabergoline/bromocriptine) STEP 1	Surgery (transssphenoidal), medical therapy STEP 1			
Parkinson disease <i>the fading substantia nigra</i> MIDBRAIN	Adult onset : 1% of people over 60, 4% over 80	Substantia nigra pars compacta (tegmentum), working with the basal ganglia	Degeneration begins with dopaminergic neurons of the SN pars compacta (the nigra loses its dark pigment)	Motor (tremor, hypokinesia); cognition (memory, executive, language); mood ; sensory (especially smell); sleep	TRAP : resting tremor, cogwheel rigidity, akinesia/bradykinesia, postural instability; Lewy bodies (α-synuclein) STEP 1	L-dopa + carbidopa STEP 1 . SN damage → motor and non-motor deficits			
Pontine stroke <i>the bilateral weakness</i> PONS	Stroke risk factors (HTN, atherosclerosis) STEP 1	Pons (supplied by basilar short circumferential branches — HNSS17)	Loss of blood supply → death of pontine nuclei and tracts	Various deficits from pontine nuclei (CN V–VIII): double vision , dizziness, balance problems, numbness, slurred speech, and weakness on BOTH sides of the body	Basilar occlusion → locked-in syndrome (quadriplegia, only vertical eye movements) STEP 1	Stroke care; reperfusion STEP 1			
Medulloblastoma <i>the child's cerebellar tumor</i> CEREBELLUM	Children : a common primary malignant brain tumor — 14.5% of new peditatric brain tumors	Cerebellum (often vermis, near the 4th ventricle STEP 1)	Primary malignant tumor; can block the 4th ventricle → obstructive hydrocephalus	Problems with walking , balance , motor skills and/or cognition ; if hydrocephalus: headache , vomiting , sleepiness	Small round blue cells, Homer-Wright rosettes ; seeds the CSF ("drop metastases") STEP 1	Surgery + chemotherapy ; ~70% 5-year relative survival			

★ **USMLE high-yield**: One disorder per region: **Broca + Huntington** (telencephalon), **pituitary adenoma** (diencephalon), **Parkinson** (midbrain), **pontine stroke** (pons), **medulloblastoma** (cerebellum). Pituitary adenoma → **bitemporal hemianopsia** + Cushing (ACTH). Parkinson = **SN pars compacta dopamine** loss. Pontine stroke → weakness on **both** sides. Medulloblastoma = child, cerebellum, 4th-ventricle hydrocephalus.

Principles of Brain Dysfunction

The four principles (apply them to every disorder)

#	PRINCIPLE EXAM EMPHASIS	LECTURE DISORDER THAT SHOWS IT	QUICK EXAMPLE
1	Damage to different structures often impacts different functions	Broca's area → language; SN → movement; pituitary → vision + hormones	Auditory cortex → hearing loss; visual cortex → vision loss
2	Which structure or pathway is damaged matters more than what causes the damage	Broca's aphasia looks the same from stroke, tumor or trauma	Stroke vs gunshot to the same visual cortex → same deficit
3	Damage to a given structure often affects more than one function	Huntington (caudate): chorea + memory + executive + language; Parkinson (SN): motor + cognition + mood + smell + sleep	—
4	Damage to distinct structures can impact the same function: networks	Gray matter regions + the white matter tracts that join them	Retina or visual cortex damage → blindness

Region → disorder map

REGION	DISORDER	CORE PICTURE
Telencephalon	Broca's aphasia · Huntington	Language loss · chorea + cognitive decline
Diencephalon	Pituitary adenoma	Bitemporal hemianopsia · Cushing (ACTH)
Midbrain	Parkinson disease	Dopamine loss in SN → tremor, hypokinesia
Pons	Pontine stroke	Cranial nerve signs + weakness on both sides
Cerebellum	Medulloblastoma	Child with ataxia ± hydrocephalus

How to use the principles on a vignette

1) **Localize first**: which structure or pathway explains **all** the deficits? 2) Then ask about the cause (stroke, tumor, degeneration, trauma) — it changes treatment, not the deficit pattern. 3) Expect **several functions** from one structure (basal ganglia and substantia nigra are motor **and** cognitive). 4) If two different lesions give the same deficit, think **networks** (gray matter + white matter tracts).

Brain networks: different structures work together to support a function, so damage anywhere in the network — even outside the brain, in the wider nervous system — can affect that function.

★ **USMLE high-yield**: Memorize the four principles word for word **EXAM EMPHASIS**: **different structures → different functions; where > what; one structure → many functions; many structures → one function (networks)**. Step 1 neuro vignettes are localization puzzles: e.g., cranial nerve palsy on one side + body weakness on the other = brainstem **STEP 1**.